

Oskar Boesch

MSc Neuro-X, EPFL
Medical Research Assistant, CHUV

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Oskar Boesch
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SKILLS

Python, C++, Swift, Java, LaTeX, Git, SQL Wetlab experience
PyTorch, nilearn, FreeSurfer, fMRIPrep, FSL French (C2), English (C1), German (B2)

RELEVANT EXPERIENCE

- Present** • **Medical Research Assistant** Leenaards Memory Centre, CHUV
– Aug 2026
Refactoring & updating the structural and functional MRI analysis pipeline for HPC clusters and applying it to IAM4HA, a study of the therapeutic effects of art and music on brain ageing (cortical thickness, functional connectivity, white-matter hyperintensities).
Bash, Python, FreeSurfer, CAT12, fMRIPrep, FSL, MARS, SHIVA
- Jun 2025** • **Startup Team Member** Innovation Booster – Digital Health Nation
– Apr 2025
Start-up team building an AI-assisted tool for rehabilitation movement-data acquisition, visualisation and analysis. Led stakeholder pitches and designed the presentation decks and posters; contributed to the Android/Flutter front-end.
Figma, Android Studio, Flutter
- Jul 2024** • **Research Engineer Intern** Lake Lucerne Institute
– Jun 2024
Built a low-cost tablet app for at-home upper-limb movement assessment, using PosePipe (R. James Cotton) for markerless pose estimation. Worked directly with clinicians and patients to refine the app.
Android Studio, PosePipe

EDUCATION

- Jul 2026** • **MSc in Neuro-X** École Polytechnique Fédérale de Lausanne
– Sept 2024
Average grade 5.5/6. Thesis supervised by Prof. Alexander Mathis, graded 6.0/6 (see Selected Projects).
- Jul 2024** • **BSc in Life Sciences Engineering** École Polytechnique Fédérale de Lausanne
– Sep 2021
Average grade 5.38/6, including an exchange year at DTU (Denmark).
- Jun 2020** • **Collège Claparède, High School**
– Aug 2016
Bilingual Maturity in English; advanced Maths and Physics, astrophysics option; incl. 6-month exchange at Cairns SHS (Australia).

SELECTED PROJECTS

- **Uncovering Hierarchical Structure in Long-Term Natural Rodent Behaviour** • Master's Thesis
Mathis Lab (EPFL), graded 6.0/6. Extended the hBehaveMAE autoencoder with layer-wise nested dropout and YaRN embeddings to learn multi-timescale mouse behaviour representations, discriminating between 80 strains.
Award Neuro-X Prize for the best Master's project in Computational Neuroscience
- **The Role of the Hippocampus in Tracking Personal Goals** • Semester Project
MIP-Lab (EPFL). Used fMRI RSA and GLM (nilearn) on 28 participants to test whether the hippocampus encodes a spatio-temporal event map, replicating Deuker et al. (2016). Found no significant effect, and identified a methodological confound in the original study.
- **Data-Driven Modeling for Robotic Mechanical Thrombectomy** • Semester Project
TRES Lab (EPFL). Built a GRU model on force-sensor and imaging data for real-time clot-length estimation in robotic thrombectomy: 76.2%/61.9% clot-start/end detection accuracy (1.5 mm margin) at 0.53 ms inference.

ADDITIONAL INFORMATION

- **Music** – Prize for both composition and performance (Collège Claparède); musical improvisation training at AMR, Geneva (Sept 2020–Jul 2021); private guitar teacher for over 5 years.
- **Swiss Military Service** (Jun 2020–Sep 2020) – Awarded athletic distinction; recommended for promotion; declined to pursue university studies.
- **Interests** – Climbing, cycling, skiing.